

Internet of Things and Wireless Networks

Project

Group No. 7

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Motivation

- Many modern homes contain at least 3 IoT capable smart home devices.
- Some devices may want to know if a user is near them.
- Even better, services might want to know where a user is and moves.
 - Wake/suspend devices which are in proximity of a user
- Privacy perspective: Can IoT devices secretly pinpoint the exact position and velocity of users?
 - This information might be very interesting for 3rd party advertisers, AI models, or intelligence (Palantir or BND).

Goal: Pinpoint the position of a Bluetooth device using triangulation.

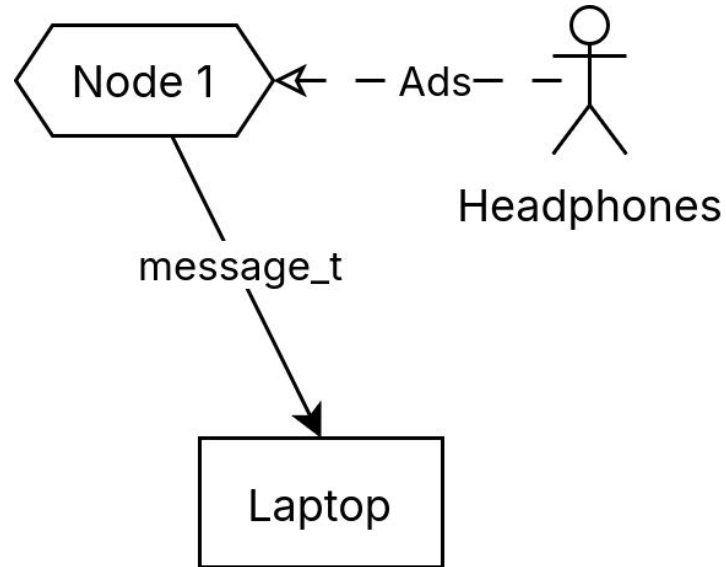
Tasks

5 Points: Setup one device which relays RSSIs of foreign advertisements to a general purpose computer to convert to distance.

10 Points: Setup three devices which relay the RSSIs of one device to a general purpose computer and triangulate the position.

15 Points: Visualize position change of a device in the network.

Solution for Task 1



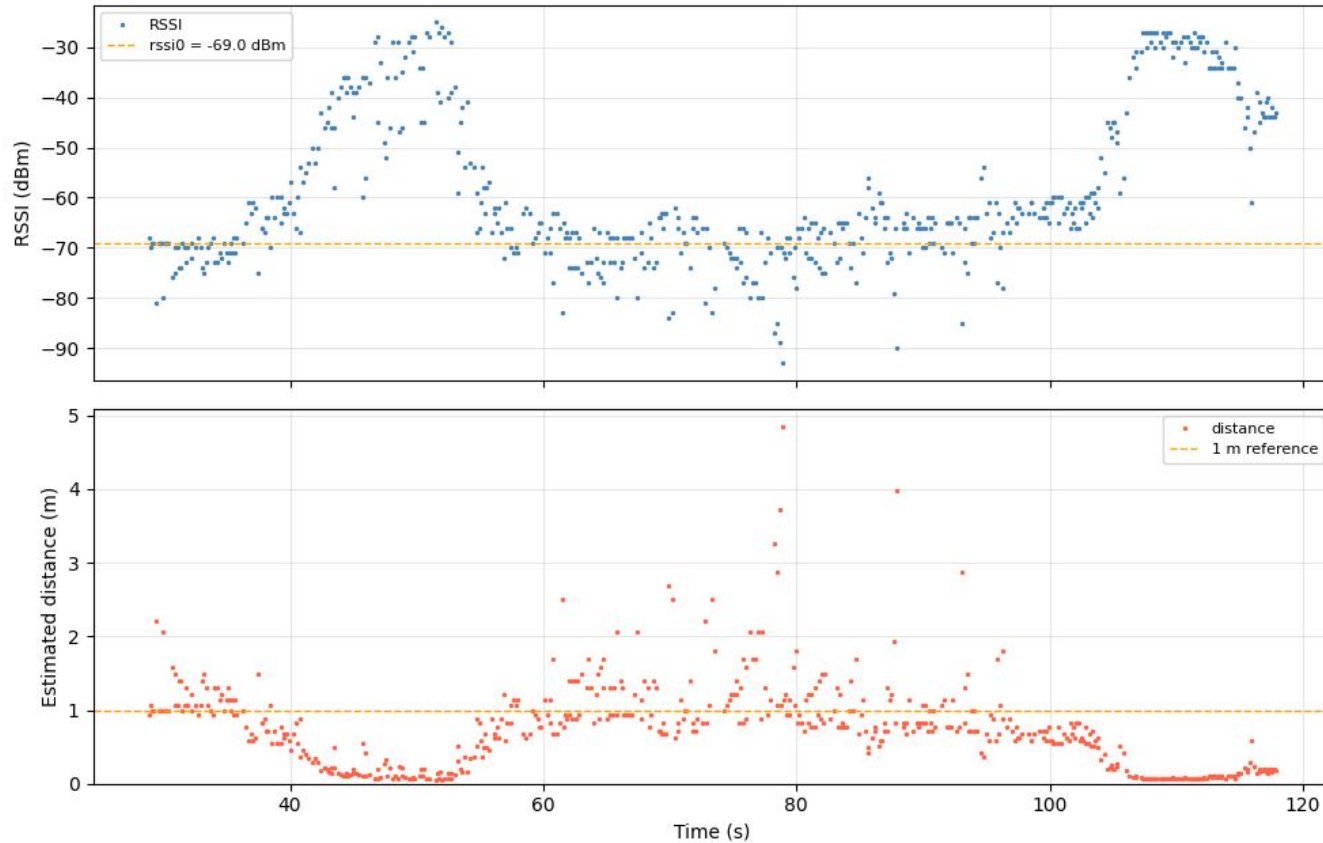
Solution for Task 1

```
1 typedef struct __packed
2 {
3     uint8_t nodeID;           // Our node ID. Either 1, 2, or 3.
4     int64_t timestamp;        // Used by the peripheral to calculate accurate positioning.
5     int8_t rssi;              // RSSI of the traffic we received.
6     bt_addr_le_t sender;      // Address of the device which sent the traffic.
7 } message_t;
```

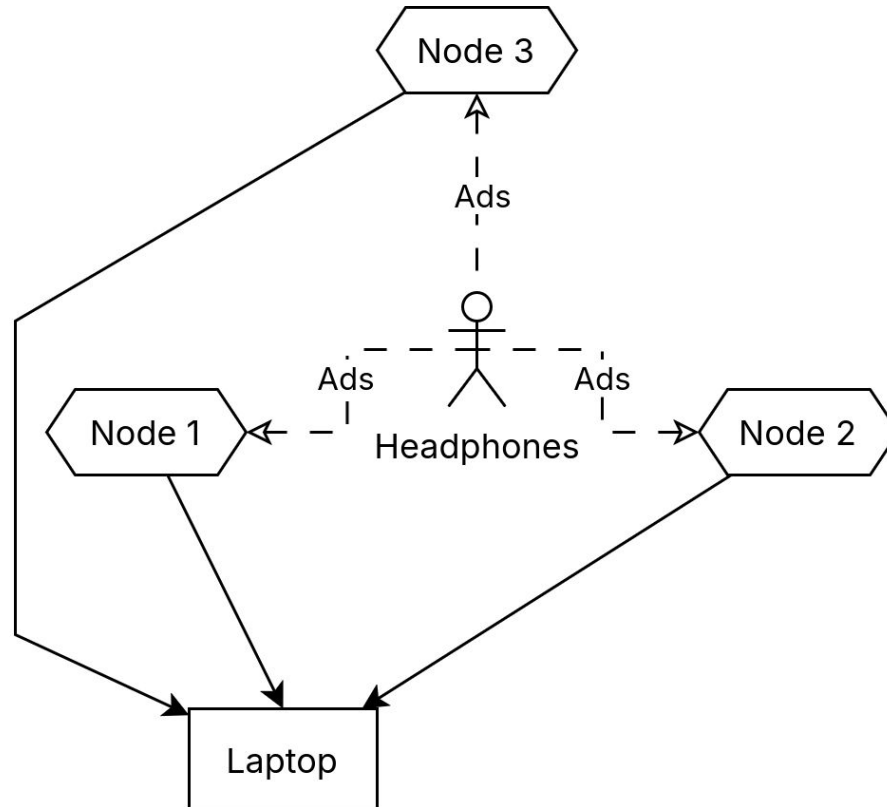
Solution for Task 1

$$d = 10^{\frac{R_0 - R}{10N}}$$

Solution for Task 1



Solution for Task 2



Solution for Task 2: Three possible approaches

- Use distance estimation of Task 1 and Barycentric coordinates
 - Unreliable, have to configure a lot of RSSI_0 values.
- Solve a very big and complicated formula
 - Maybe too slow for my laptop
- Approximate the formula using footprinting
 - Has some configuration overhead, but else very fast.

Solution for Task 2: Formulas

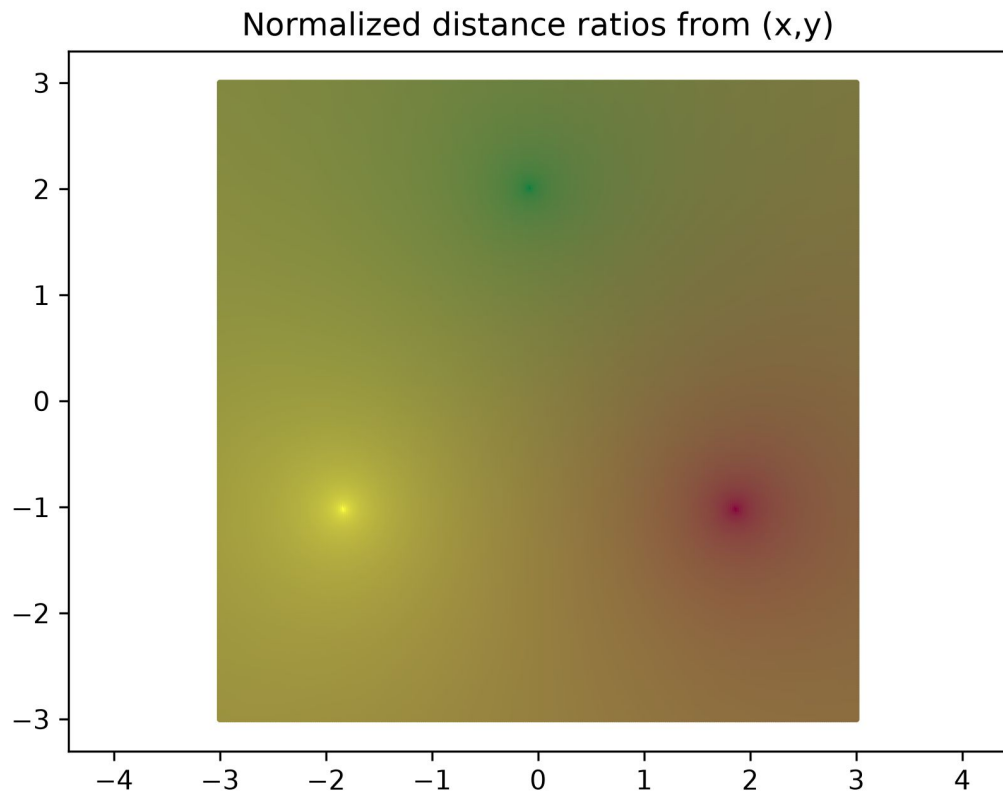
$$R_i = R_{0_i} - 10N \cdot \log_{10}(d_i)$$

$$\begin{aligned} R_i - R_j &= R_0 - R_0 - 10N \cdot \log_{10}(d_i) + 10N \cdot \log_{10}(d_j) \\ &= 10N \cdot \log_{10}(d_j/d_i) \end{aligned}$$

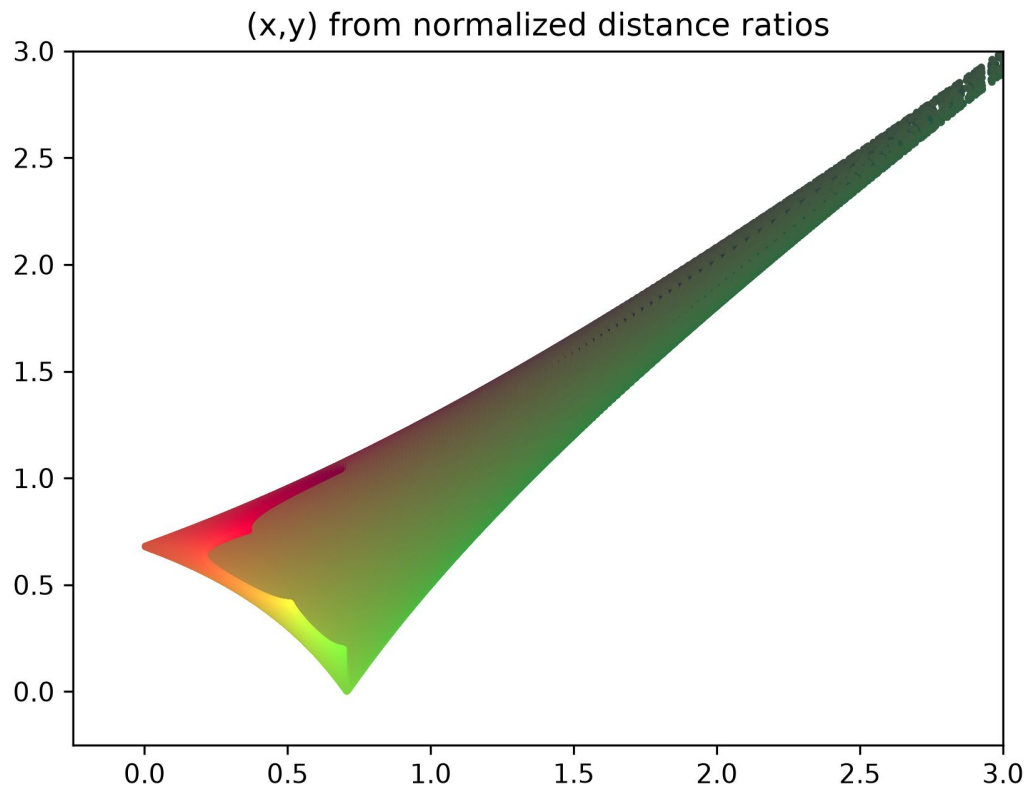
$$d_j/d_i = 10^{\frac{R_i - R_j}{10N}}$$

Idea for Approaches:
Calculate position
from distance ratio!

Solution for Task 2: k-d Tree Position Lookup



Solution for Task 2: k-d Tree Position Lookup



Solution for Task 3: Triangulation Visualizer

Demo

Thanks for listening!